Worksheet 1: Orbits and Formation

Covers Lectures 1 and 2: history of the field, planet formation, orbital dynamics

Planetary Systems (WBAS002-05) • Kapteyn Astronomical Institute, University of Groningen

This worksheet is ungraded practice, with the same shape and level as the final exam. Plan up to 2 hours for a serious pass. Work each problem through on paper before you open the solutions, and show your algebra: the exam asks for the steps, not only the number. Some parts state an intermediate result ("show that…"); use it to check your work, and to continue if a step resists. Carry at least four significant figures through the intermediate steps (the worked solutions print five) and round only at the end.

Constants and data

$G = 6.674\times10^{-11}\ \mathrm{m^3\,kg^{-1}\,s^{-2}}$ $\quad$ 1 AU = 1.496×10^{11} m $\quad$ 1 yr = 3.156×10^{7} s $\quad$ 1 d = 86 400 s
$M_\odot = 1.989\times10^{30}$ kg $\quad$ $M_\oplus = 5.972\times10^{24}$ kg $\quad$ $R_\oplus$ = 6371 km

Body	a	P	Mass ($M_\oplus$)	ρ (kg m^{-3})
Mercury	0.387 AU	0.241 yr	0.055	5427
Venus	0.723 AU	0.615 yr	0.815	5243
Earth	1.000 AU	1.000 yr	1.000	5514
Mars	1.524 AU	1.881 yr	0.107	3934
Jupiter	5.203 AU	11.86 yr	317.8	1326
Saturn	9.537 AU	29.46 yr	95.16	687
Uranus	19.19 AU	84.01 yr	14.54	1271
Neptune	30.07 AU	164.8 yr	17.15	1638
Moon	3.844×10^{5} km	27.32 d	0.0123	3344
Io		1.769 d		
Europa		3.551 d		
Ganymede	1.0704×10^{6} km	7.155 d		

Saturn's mean radius is 58 232 km. Planetary and satellite values from NASA/JPL Solar System Dynamics; Galilean satellite periods as quoted in the Lecture 2 notes.

Problem 1 Weighing the Sun and Jupiter

Newton's form of Kepler's third law turns a period and an orbital size into a mass. Here you derive the equation, test a claim against it, and then weigh the Sun with it.

(a) A planet of mass m moves on a circular orbit of radius r about a star of mass M, with $m \ll M$. Starting from the balance between the gravitational force and the centripetal force, derive

$$M = \frac{4\pi^2 r^3}{GP^2}.$$

(b) Judge this claim in 2 to 3 sentences: *"If Jupiter were twice as massive, its orbital period at its present distance would be noticeably longer."*

(c) Compute the mass of the Sun from Jupiter's orbit (a = 5.203 AU, P = 11.86 yr; show first that $r = 7.784 \times 10^{11}$ m and $P = 3.743 \times 10^{8}$ s).

Problem 2 The cheapest route to Mars

Treat the orbits of Earth and Mars as circular and coplanar, with radii 1.000 AU and 1.524 AU. A spacecraft transfers between them on an ellipse whose perihelion touches Earth's orbit and whose aphelion touches Mars's orbit. At each end the spacecraft fires its engine briefly (a *burn*), changing its speed by an increment Δv; a burn along the direction of motion is *prograde*. Work entirely in the Sun's gravitational field and ignore the gravity of Earth and Mars themselves.

(a) Two spacecraft of equal mass are on orbits with the same semi-major axis a but different eccentricities. Judge each claim, with justification: (i) *"they have the same total energy"*; (ii) *"they pass through the distance $r = a$ at the same speed"*.

(b) Compute the semi-major axis a_t of the transfer ellipse. Explain why the flight time is half the period of that ellipse, and compute it in days.

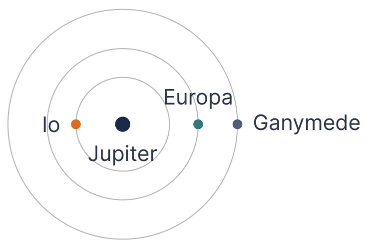
(c) Use the vis-viva equation to show that the spacecraft moves at $v_p = 32.7\ \mathrm{km\,s^{-1}}$ at the transfer perihelion and $v_a = 21.5\ \mathrm{km\,s^{-1}}$ at the transfer aphelion. The circular orbital speeds, from the $r = a$ case of part (a), are $v_\oplus = 29.79\ \mathrm{km\,s^{-1}}$ and $v_{\mathrm{Mars}} = 24.13\ \mathrm{km\,s^{-1}}$. From the four speeds, form the two velocity increments Δv_1 (departure), Δv_2 (arrival), and their sum, and explain why *both* burns point in the direction of motion.

Problem 3 The Laplace resonance of the Galilean moons

Lecture 2 describes the orbital periods of Io, Europa and Ganymede as standing in a 1 : 2 : 4 ratio. This problem asks how exact that statement is, and what the resonance geometry enforces.

(a) Compute the two consecutive period ratios and the Ganymede-to-Io ratio, and give the percentage by which each misses its integer value.

(b) The figure shows the three orbits at the moment of a Europa–Ganymede conjunction; the resonance locks the phases such that Io then stands on the *opposite* side of Jupiter.



Idealise the periods as exactly 1 : 2 : 4. Sketch the positions of the three moons after 1, 2, and 4 Io periods. From your sketch: (i) where does Io stand at every Europa–Ganymede conjunction, and (ii) can a triple conjunction occur?

(c) The resonance holds Io's eccentricity at $e \approx 0.004$ rather than letting tides circularise the orbit. Explain why a non-zero eccentricity is needed for tidal heating to continue in a moon that is already tidally locked, and name the observable consequence.

Problem 4 Tides and the Roche limit

(a) A moon of radius R_s orbits at distance d from a planet of mass M_p, with $R_s \ll d$. Show that the difference between the planet's gravitational acceleration at the moon's near surface and at its centre is

$$\Delta a \approx \frac{2GM_pR_s}{d^3}.$$

Use $(1 - x)^{-2} \approx 1 + 2x$ for $x \ll 1$.

(b) The fluid Roche limit is

$$d_R \approx 2.46\, R_p \left(\frac{\rho_p}{\rho_s}\right)^{1/3}.$$

Interrogate this formula before using it: (i) why does the *satellite's* radius R_s not appear? (ii) what does it predict as $\rho_s \to \infty$, and does that make sense? (iii) what does it predict for $\rho_s = \rho_p$, and what does that say about how close *any* self-gravitating satellite can orbit?

(c) Evaluate d_R for Saturn (R_p = 58 232 km, ρ_p = 687 kg m^{-3}) and a satellite of solid water ice (ρ_s = 1000 kg m^{-3}). Saturn's main rings span 67 000 to 137 000 km from the planet's centre, with the F ring near 140 000 km: compare. Ring particles are in fact porous aggregates with $\rho_s \approx$ 500 kg m^{-3}; *without* recomputing from scratch, state the factor by which the limit moves and what that does to your comparison.

Problem 5 From planetesimals to planets

(a) Gravitational focusing raises the collision cross-section of a body of radius R to

$$\sigma_{\rm eff} = \pi R^2 \left(1 + \frac{v_{\rm esc}^2}{v_\infty^2}\right).$$

Two planetesimals of density 3000 kg m^{-3} have radii 50 km and 500 km. Compute the escape speed of the smaller body, and explain why $v_{\rm esc} \propto R$ at fixed density. Then compute both focusing factors for a swarm with random velocity v_∞ = 10 m s^{-1}, and compare the ratio of the two effective cross-sections with the ratio of the geometric ones.

(b) Read the limits of the focusing formula: what does $\sigma_{\rm eff}$ reduce to for $v_\infty \gg v_{\rm esc}$, and how does it scale with R for $v_\infty \ll v_{\rm esc}$? Confirm the first limit by evaluating both focusing factors at v_∞ = 500 m s^{-1}. The growing bodies themselves stir the swarm toward their own escape speed: name the two growth regimes these limits correspond to, and explain why the first regime shuts itself down.